



SOICT

HUST

ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

Multimodal Machine Learning

Lecture 1.2: Multimodal Research Tasks

Đàm Quang Tuấn

Lecture Objectives

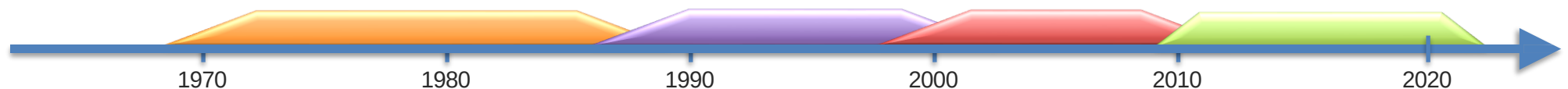
- A historical view on multimodal research
- Experimental design
 - Research questions and hypotheses
- Multimodal datasets and research tasks
 - 100+ multimodal datasets (+ curated list)
- Examples of previous course projects

Multimodal Research: A Historical View

Prior Research in “Multimodal”

Four eras of multimodal research

- The “behavioral” era (1970s until late 1980s)
- The “computational” era (late 1980s until 2000)
- The “interaction” era (2000 - 2010)
- The “deep learning” era (2010s until ...)
 - ❖ Main focus of this course



Behavioral Study of Multimodal



Language
and gestures

David McNeill

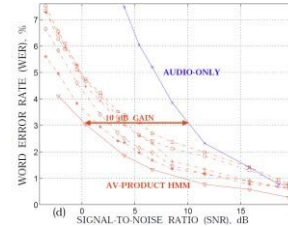
“For McNeill, gestures are in effect the speaker’s thought in action, and integral components of speech, not merely accompaniments or additions.”

McGurk effect



➤ The “Computational” Era (Late 1980s until 2000)

1) Audio-Visual Speech Recognition



Redundancy between audio and visual modalities help with handling noise and with robustness

2) Multimodal interfaces



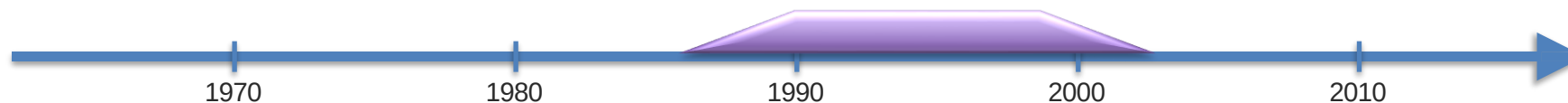
Affective Computing is computing that relates to, arises from, or deliberately influences emotion or other affective phenomena.

3) Multimedia



[1994-2010]

“...automatically combines speech, image and natural language understanding to create a full-content searchable digital video library.”



➤ The “Interaction” Era (2000s)

Modeling Multimodal Social Interactions



AMI Project [2001-2006, IDIAP]

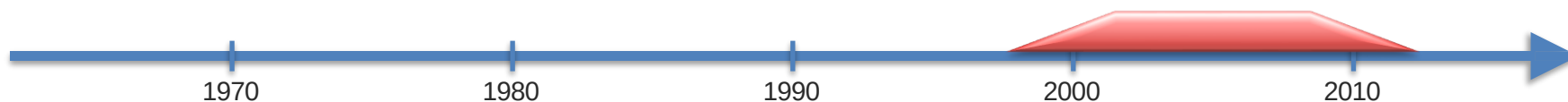
- 100+ hours of meeting recordings
- Transcribed and annotated

CALO Project [2003-2008, SRI]

- Cognitive Assistant that Learns and Organizes
- Siri was a spinoff from this project

SSP Project [2008-2011, IDIAP]

- Social Signal Processing
- Great dataset repository: <http://sspnet.eu/>



➤ The “deep learning” era (2010s until ...)

Representation learning (a.k.a. deep learning)

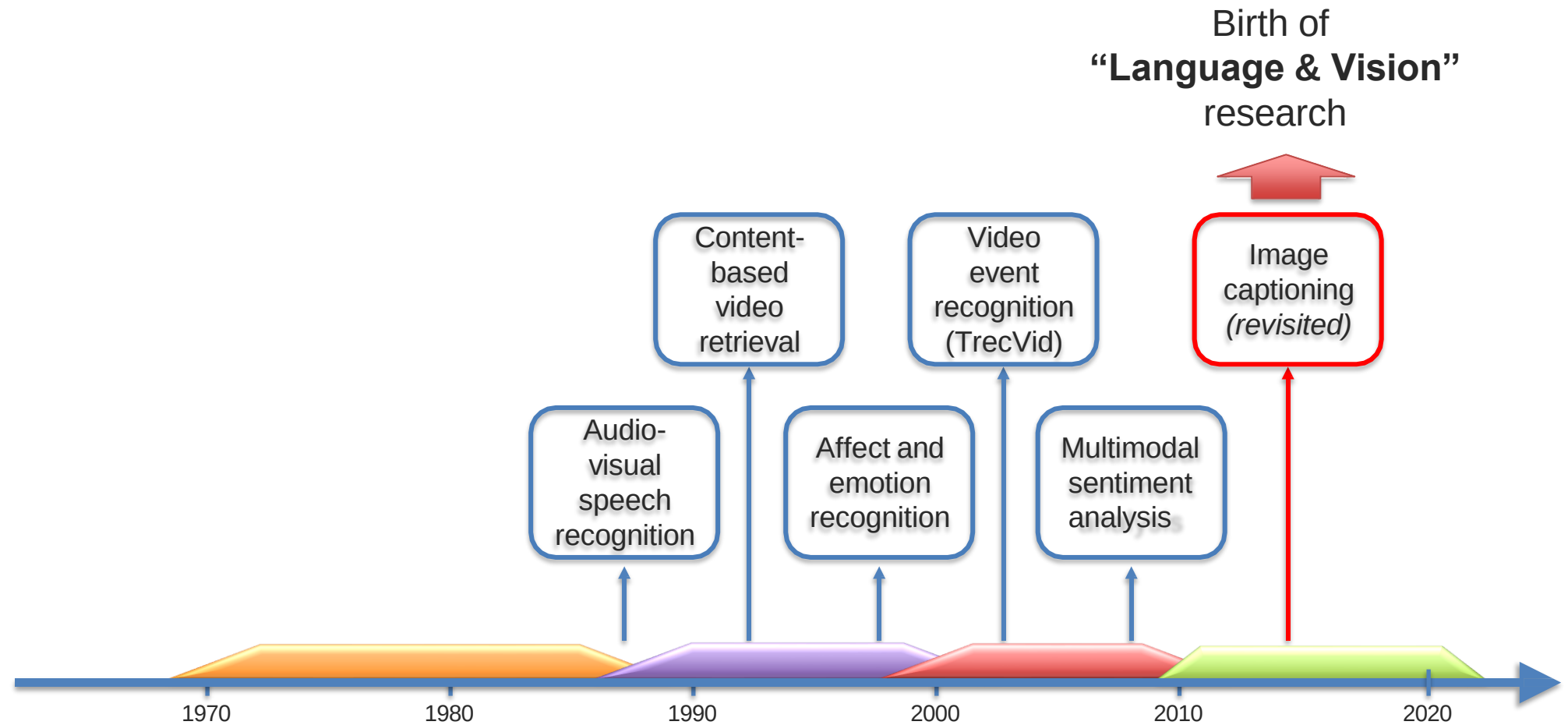
- Multimodal deep learning [ICML 2011]
- Multimodal Learning with Deep Boltzmann Machines [NIPS 2012]
- Visual attention: Show, Attend and Tell: Neural Image Caption Generation with Visual Attention [ICML 2015]

Key enablers for multimodal research:

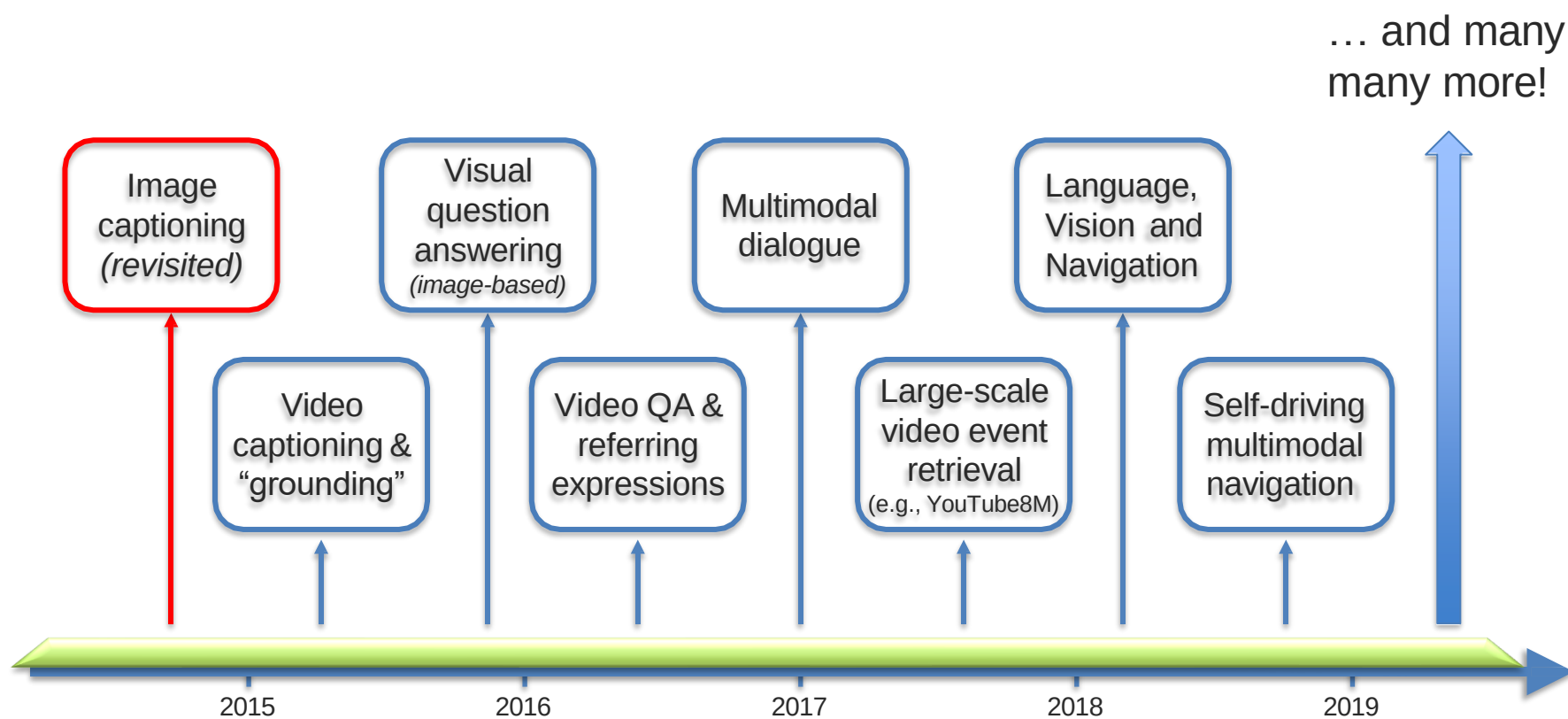
- New large-scale multimodal datasets
- Faster computer and GPUS
- High-level visual features
- “Dimensional” linguistic features



Multimodal Research Tasks



Multimodal Research Tasks



Real world tasks tackled by Multimodal ML

A. Affect recognition

- Emotion
- Personalities
- Sentiment



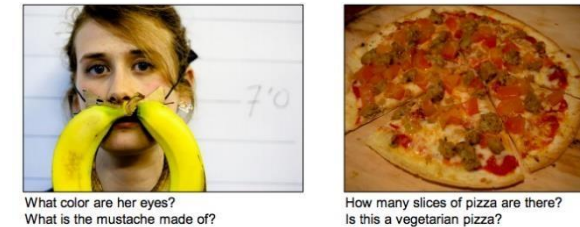
B. Media description

- Image and video captioning



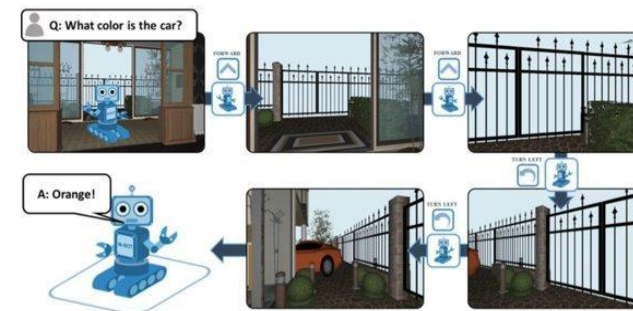
C. Multimodal QA

- Image and video QA
- Visual reasoning



D. Multimodal Navigation

- Language guided navigation
- Autonomous driving



Real world tasks tackled by Multimodal ML

E. Multimodal Dialog

- Grounded dialog

F. Event recognition

- Action recognition
- Segmentation

G. Multimedia information retrieval

- Content based/Cross-media



(a) get-out-car



(a) fight-person



(b) push-up



(b) cartwheel



Multimodal Datasets

Our Latest List of Multimodal Datasets

A. Affect Recognition

AFEW	A1	1
AVEC	A2	1
IEMOCAP	A3	1
POM	A4	3
MOSI	A5	3
CMU-MOSEI	A6	3
TUMBLR	A7	2
AMHUSE	A8	1
VGD	A9	3
Social-IQ	A10	3
MELD	A11	3
MUSTARD	A12	3
DEAP	A13	3
MAHNOB	A14	3
Continuous LIRIS-ACCEDE	A15	2
DECAF	A16	2
ASCERTAIN	A17	2
AMIGOS	A18	2
EMOTIC	A19	3
M2H2	A20	3
UR-Funny	A21	3
CH-SIMS	A22	3
MuSe-CaR	A23	2
MEmoR	A24	2

B. Media Description

MSCOCO	B1	1
MPII	B2	2
MONTREAL	B3	2
LSMDC	B4	2
CHARADES	B5	3
REFEXP	B6	3
GUESSWHAT	B7	3
FLICKR30K	B8	1
CSI	B9	1
MIT-MIT	B10	3
MVSQ	B11	2
NeuralWalker	B12	2
Visual Relation	B13	3
Visual Genome	B14	3
Pinterest	B15	2
Movie Graph	B16	3
nocaps	B17	3
CrossTask	B18	2
Refer360	B19	3
Towers of Babel (WikiScenes)	B20	3
N24News	B21	2
Localized Narratives	B22	3

Our Latest List of Multimodal Datasets

1	Good reference dataset, but maybe not as suited for a course project	
2	It may be a good idea, but also look at alternatives.	
3	Usually more recent datasets, well-suited for the project	

C. Multimodal QA

VQA	C1	1
DAQUAR	C2	1
COCO-QA	C3	2
MADLIBS	C4	2
TEXTBOOK	C5	3
VISUAL7W	C6	3
TVQA	C7	3
VCR	C8	3
Cornell NLVR	C9	3
Cornell NLVR2	C10	3
CLEVR	C11	3
EQA	C12	3
TextVQA	C13	3
GQA	C14	3
CompGuessWhat	C15	3
DVD	C16	2
AGQA	C17	3
VizWiz	C18	3
SUTD-TrafficQA	C19	3
WebQA	C20	3

D. Multimodal Navigation

Room-2-Room (R2R)	D1	1
RERERE	D2	2
VNLA	D3	3
nuScenese	D4	3
Waymo	D5	3
CARLA	D6	1
Argoverse	D7	3
ALFRED	D8	2
TEACH	D9	2
Room-across-room (RxR)	D10	3
Winoground	D11	3

Our Latest List of Multimodal Datasets

1 Good reference dataset, but maybe not as suited for a course project

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3 Usually more recent datasets, well-suited for the project

E. Multimodal Dialog

VISDIAL	E1	3
Talk the Walk	E2	3
Vision-and-Dialog Navigation	E3	3
CLEVR-Dialog	E4	2
Fashion Retrieval	E5	2
MMD	E6	1

F. Event Understanding

WHATS-COOKING	F1	1
TACOS	F2	2
TACOS-MULTI	F3	2
YOU-COOK	F4	1
MED	F5	1
TITLE-VIDEO-SUMM	F6	2
MEDIA-EVAL	F7	3
CRISMMMD	F8	3
EPIC-KITCHENS	F9	2
Fakeddit	F10	2

G. Cross-media Retrieval

IKEA	G1	3
MIRFLICKR	G2	3
NUS-WIDE	G3	1
YAHOO-FLICKR	G4	1
YOUTUBE-8M	G5	2
YOUTUBE-BOUNDING	G6	2
YOUTUBE-OPEN	G7	2
VIST	G8	3
Recipe1M+	G9	3
VATEX	G10	3

... and please let us know (via Piazza) when you find more!

A Curated List of Multimodal Datasets

- MOSEI: Sentiment and Emotion (A6)
- Social-IQ: Modeling Social Interaction (A10)
- MELD: multi-party dialogue and emotions (A11, E)
- TVQA: Video Understanding (C7)
- NLVR2: Natural Language Grounding & Reasoning (C10)
- WebQA: Multi-hop visual and text reasoning (C20)
- Room-Across-Room: Navigation (D10)
- Winoground: Compositionality (D11)
- IKEA: multimodal retrieval (G1)

But please explore other datasets as well!!

But please explore other datasets as well!!

Affect recognition dataset 2 (A2)

- Three AVEC challenge datasets 2011/2012, 2013/2014, 2015, 2016, 2017, 2018
- Audio-Visual emotion recognition
- Labeled for dimensional emotion (per frame)
- 2011/2012 has transcripts
- 2013/2014/2016 also includes depression labels per subject
- 2013/2014 reading specific text in a subset of videos
- 2015/2016 includes physiological data
- 2017/2018 includes depression/bipolar



[AVEC 2011/2012](#)



[AVEC 2013/2014](#)



[AVEC 2015/2016](#)

Multimodal Sentiment Analysis (A6)

- Multimodal sentiment and emotion recognition
- [CMU-MOSEI](#) : 23,453 annotated video segments from 1,000 distinct speakers and 250 topics

*And he I don't think he got mad when hah
I don't know maybe.*

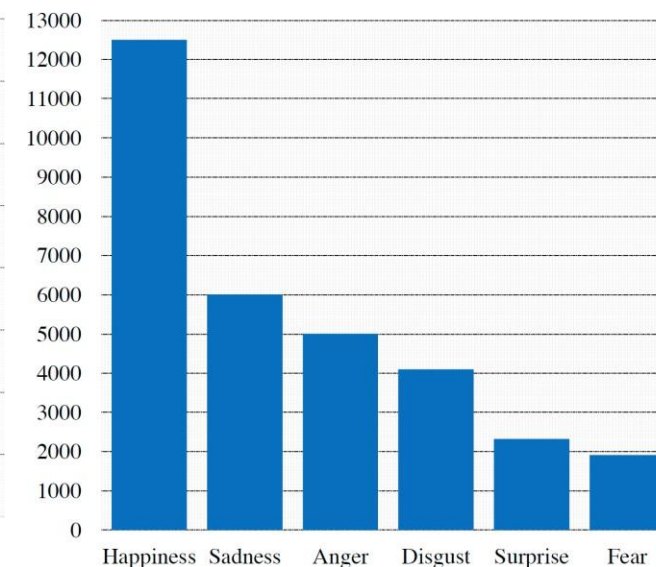
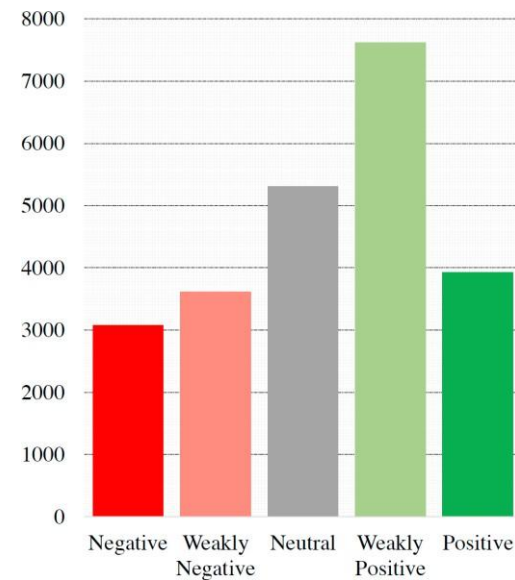


(frustrated voice)

All I can say is he's a pretty average guy.



(disappointed voice)



Media description dataset 1 – MS COCO (B1)

- Microsoft Common Objects in COntext ([MS COCO](#))
- 120000 images
- Each image is accompanied with five free form sentences describing it (at least 8 words)
- Sentences collected using crowdsourcing (Mechanical Turk)
- Also contains object detections, boundaries and keypoints



The man at bat readies to swing at the pitch while the umpire looks on.



A large bus sitting next to a very tall building.

Visual Questions & Answers – VQA (C1)

- Task - Given an image and a question, answer the question (<http://www.visualqa.org/>)



What color are her eyes?
What is the mustache made of?



How many slices of pizza are there?
Is this a vegetarian pizza?



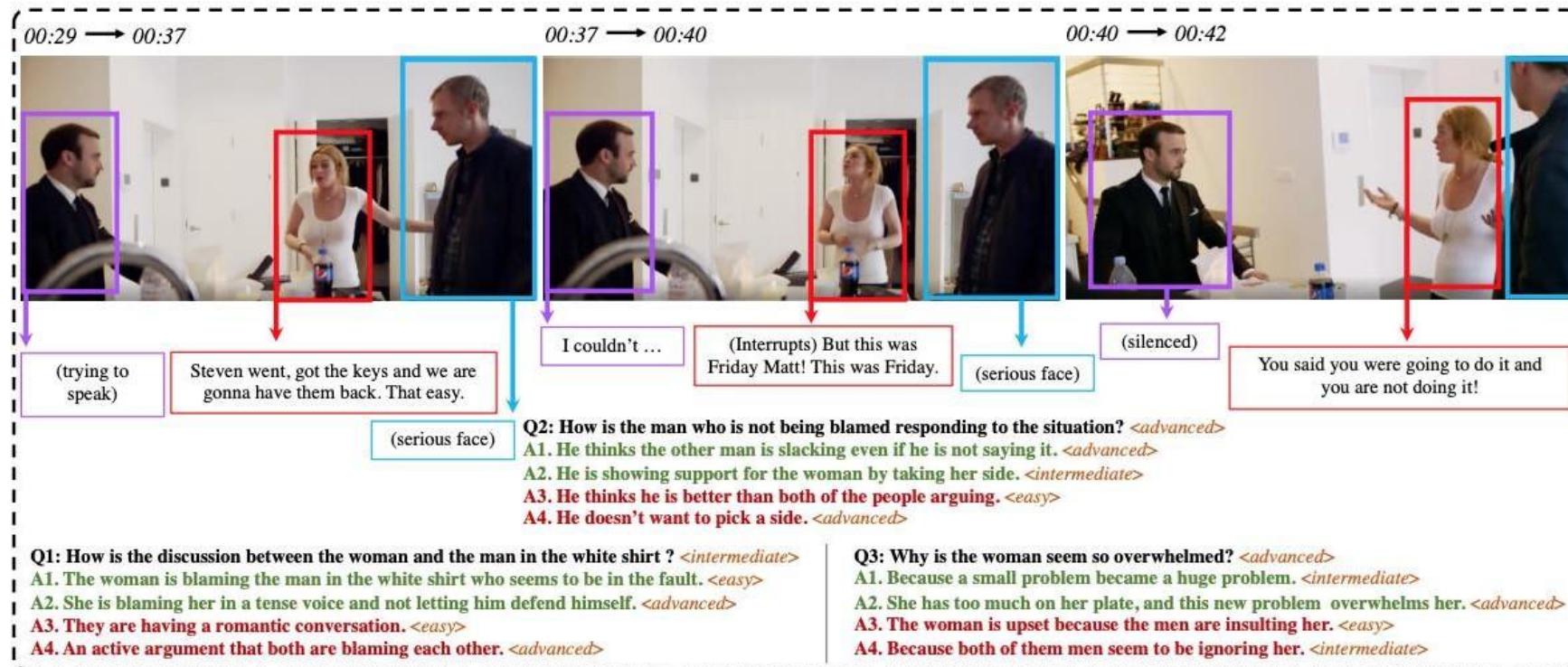
Is this person expecting company?
What is just under the tree?



Does it appear to be rainy?
Does this person have 20/20 vision?

Social Interaction Q&A Dataset (A10)

- [Social-IQ](#): 1.2k videos, 7.5k questions, 50k answers
- Questions and answers centered around social behaviors



Multimodal QA (C7)

■ TVQA

- Video QA dataset based on 6 popular TV shows
- 152.5K QA pairs from 21.8K clips
- Compositional questions

00:00.755 --> 00:02.655
(Chandler:) Go to your room!
00:06.961 --> 00:08.622
(Janice:) I gotta go, I gotta go.

00:08.829 --> 00:10.057
(Janice:) Not without a kiss.
00:10.264 --> 00:12.391
(Chandler:) Maybe I won't kiss you so you'll stay.

00:12.600 --> 00:14.761
(Joey:) Kiss her. Kiss her!
00:16.771 --> 00:19.137
(Janice:) I'll see you later, sweetie. Bye, Joey.

...

00:39.327 --> 00:40.760
(Chandler:) She makes me happy.
00:41.596 --> 00:44.087
(Joey:) Okay. All right.

...

00:00 00:06 00:10 00:17 00:39 00:45 01:04

What is Janice holding on to after Chandler sends Joey to his room?

A Chandler's tie
B Chandler's hands
C Her Breakfast
D Her coat
E Chandler's coffee cup.

Why does Joey want Chandler to kiss Janice when they are in the kitchen?

A Because Joey is glad that Chandler is happy
B Because Joey likes to watch people kiss
C Because then she will leave
D Because Joey thinks Janice is hot
E Because then Chandler will move away from the toast.

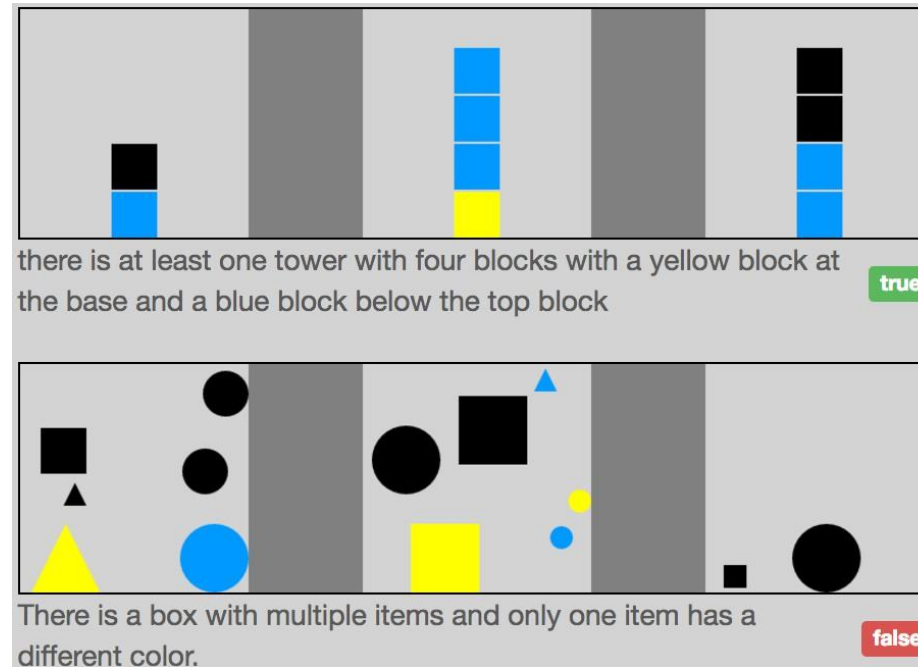
What is on the couch behind Joey when he is at the counter?

A A chick
B A soccer ball
C A duck
D A pillow
E Janice's coat

Multimodal QA – Visual Reasoning (C9)

- Cornell NLVR

- 92,244 pairs of natural language statements grounded in synthetic images
- Determine whether a sentence is true or false about an image



Multimodal QA – Visual Reasoning (C10)

- Cornell NLVR2
 - Same as NLVR but with >100k real images



The left image contains twice the number of dogs as the right image, and at least two dogs in total are standing.



One image shows exactly two brown acorns in back-to-back caps on green foliage.

WebQA (C20)

- <https://webqna.github.io/>
- Given a question Q, and a list of sources S = {s1, s2, ...}, a system must a) identify the sources from which to derive the answer, and b) generate an answer as a complete sentence.

Q: At which festival can you see a castle in the background: Oktoberfest in Domplatz Austria or Tanabata festival in Hiratsuka, Japan?

 <i>J24 029 Dom, Oktoberfest</i>	 <i>The festival is a "Syonan Hiratsuka Tanabata Matsuri".</i>	In 1938, after Hitler had annexed Austria and won the Sudetenland via the Munich Agreement, Oktoberfest was renamed to Großdeutsches Volksfest (Greater German folk festival), and as a showing of strength, the Nazi regime transported people from Sudetenland to the Wiesn by the score.	Large-scale Tanabata festivals are held in many places in Japan, mainly along shopping malls and streets, which are decorated with large, colorful streamers. The most famous Tanabata festival is held in Sendai from 6 to 8 August.	 <i>Calella - Catalonia, Spain - 11 Aug. 2009</i>	
In the summer, the Sendai Tanabata Festival, the largest Tanabata festival in Japan, is held. In winter, the trees are decorated with thousands of lights for the Pageant of Starlight, lasting through most of December.	 <i>Masskrüge Four mugs of beer at Oktoberfest 2008.</i>	 <i>Fussa Tanabata Festival-Tokyo</i>	 <i>Tanabata festival in Hiratsuka</i>	For the Oktoberfest, Löwenbräu brews a special Märzen beer called Oktoberfestbier or Wiesenbier ("meadow beer," referring to the Bavarian name of the festival site, the "Wiesn").	 <i>Ghost train on the Munich Oktoberfest.</i>

A: You can see a castle in the background at Oktoberfest in Domplatz, Austria

Navigating in a Virtual House (D1)

Visually-grounded natural language navigation in real buildings

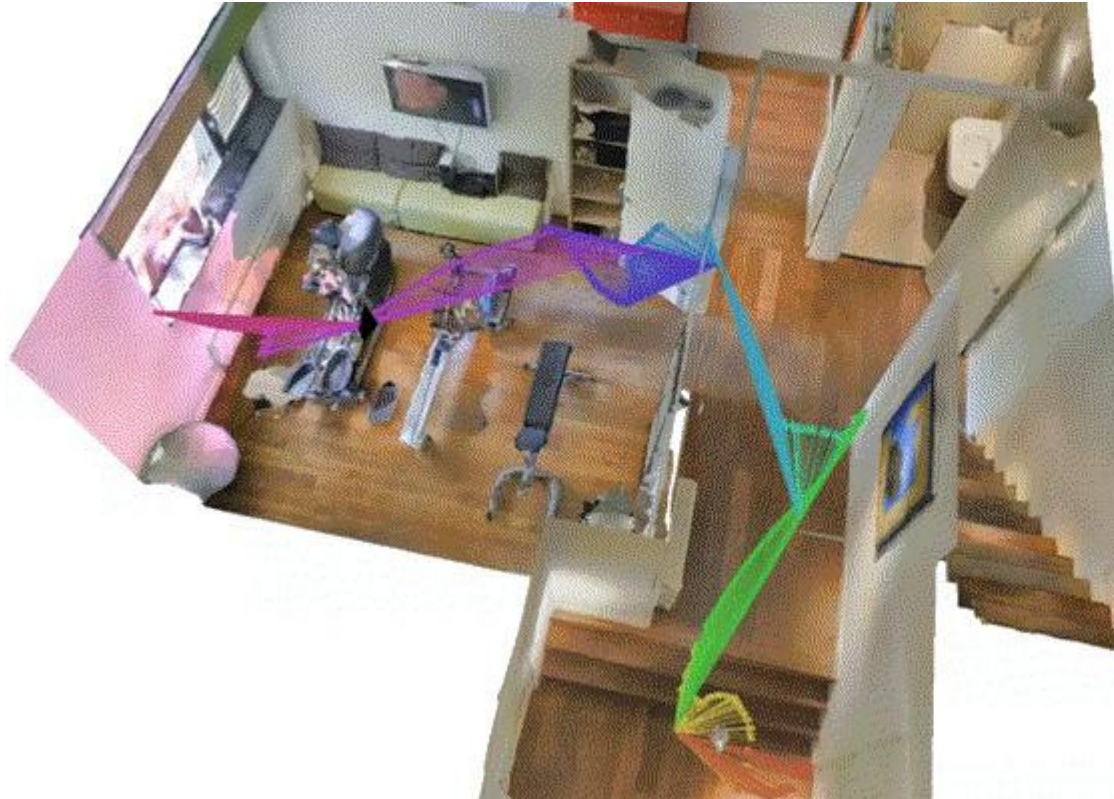
- [Room-2-Room](#): 21,567 open vocabulary, crowd-sourced navigation instructions



Instruction: Head upstairs and walk past the piano through an archway directly in front. Turn right when the hallway ends at pictures and table. Wait by the moose antlers hanging on the wall.

Room-Across-Room (D10)

- [Github](#)
- Similar to Room-to-Room (D1) except larger, multilingual, and with longer paths



Now you are standing in-front of a closed door, turn to your left, you can see two wooden steps, climb the steps and walk forward by crossing a wall paint which is to your right side, you can see open door enter into it.

This is a gym room, move forward, walk till the end of the room, you can see a grey colored ball to the corner of the room, stand there, that's your end point.

Winoground (D11)

- [Github](#)
- Same words, different order, different images. Intended to test the compositionality of vision-language models



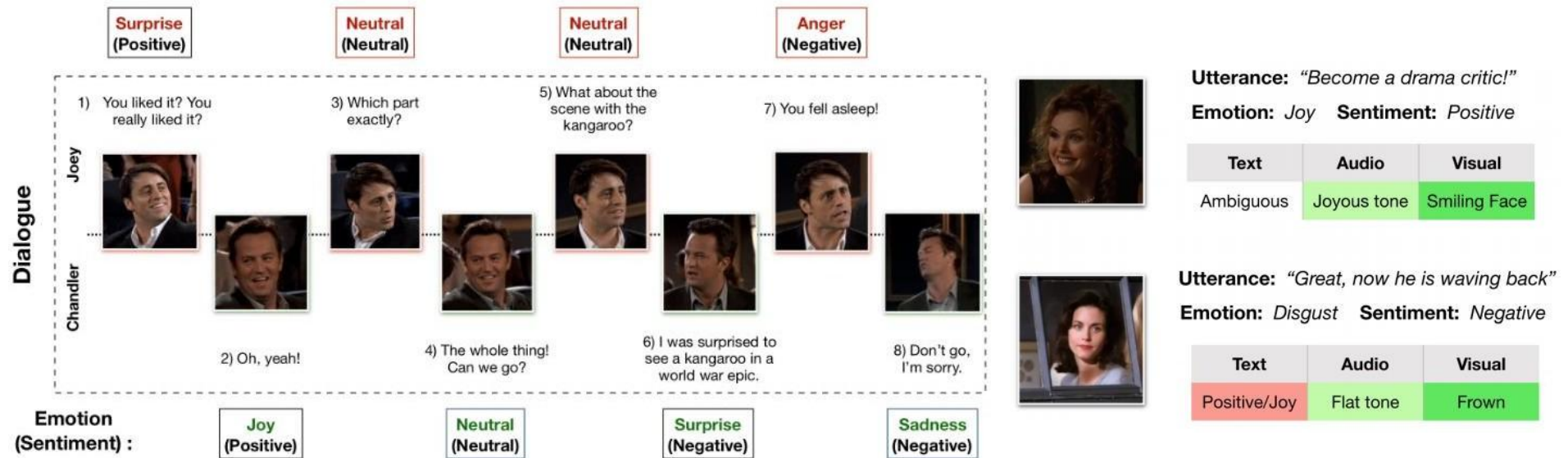
(a) some plants
surrounding a
lightbulb



(b) a lightbulb surrounding some plants

Multi-Party Emotion Recognition (A11, E)

- [MELD](#): Multi-party dataset for emotion recognition in conversations



EPIC-Kitchens (F9)

- [Dataset](#)
- Large-scale dataset in first-person (egocentric) vision; multi-faceted, audio-visual, non-scripted recordings in native environments - i.e. the wearers' homes



Multimodal Retrieval: IKEA Interior Design Dataset (G1)

- [Interior Design Dataset](#) – Retrieve desired product using room photos and text queries.
- 298 room photos, 2193 product images/descriptions.

Room images:



Object images: Description:



You sit comfortably thanks to the armrests.



There's a natural and living feeling of wood, as knots and other marks remain on the surface.



This lamp gives a pleasant light for dining and spreads a good directed light across your dining or bar table.

Some Advice About Multimodal Datasets

- If you are used to deal with text or speech
 - Space will become an issue working with image/video data
 - Some datasets are in 100s of GB (compressed)
- Memory for processing it will become an issue as well
 - Won't be able to store it all in memory
- Time to extract features and train algorithms will also become an issue
- Plan accordingly!
 - Sometimes tricky to experiment on a laptop (might need to do it on a subset of data)

Available Tools

- Use available tools in your research groups
 - Or pair up with someone that has access to them
- Find some GPUs!
- We will be getting AWS credit for some extra computational power
- Google Cloud Platform credit as well



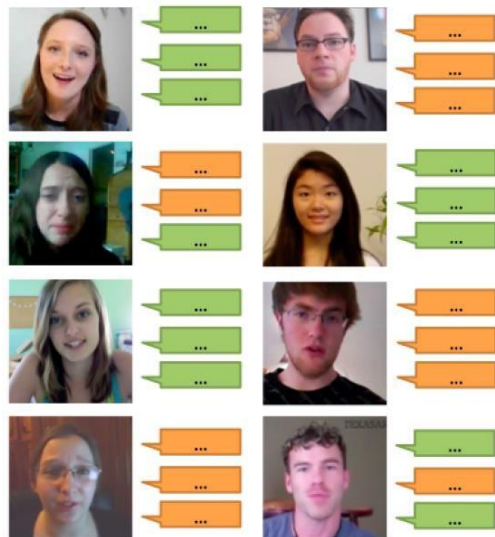
Examples of Previous Projects

Project Example: Select-Additive Learning

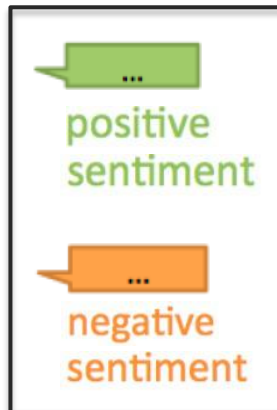
Research task: Multimodal sentiment analysis

Datasets: MOSI, YouTube, MOUD

Main idea: Reducing the effect of *confounding factors* when limited dataset size



Legend



What rules can you infer from this data?

- ✓ Smile -> positive sentiment
- ✓ Frown -> negative sentiment
- ✓ nod -> positive sentiment
- ✗ Wearing glasses -> negative sentiment

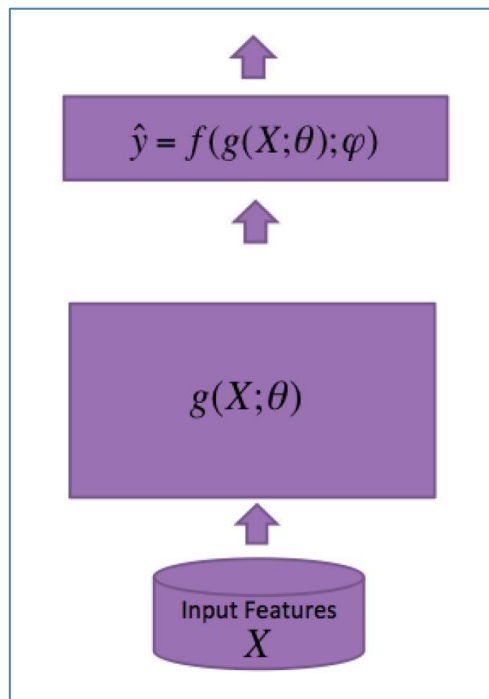
Confounding factor!

Haohan Wang, Aaksha Meghawat, Louis-Philippe Morency and Eric P. Xing, Select-additive Learning: Improving Generalization In Multimodal Sentiment Analysis, ICME 2017, <https://arxiv.org/abs/1609.05244>

Project Example: Select-Additive Learning

Solution: Learning representations that reduce the effect of user identity

“Conventional”
representation learning



Select-Additive Learning



Hypothesis: the representation is a mixture from the person-independent factor $g(X)$ and the person-dependent factor $h(Z)$.

Haohan Wang, Aaksha Meghawat, Louis-Philippe Morency and Eric P. Xing, Select-additive Learning: Improving Generalization In Multimodal Sentiment Analysis, ICME 2017, <https://arxiv.org/abs/1609.05244>

Project Example: Word-Level Gated Fusion

Research task: Multimodal sentiment analysis

Datasets: MOSI, YouTube, MOUD

Main idea: Estimating importance of each modality at the word-level in a video.



Visual Gate:

Reject

Pass

Reject



Visual modality: Hands cover mouth

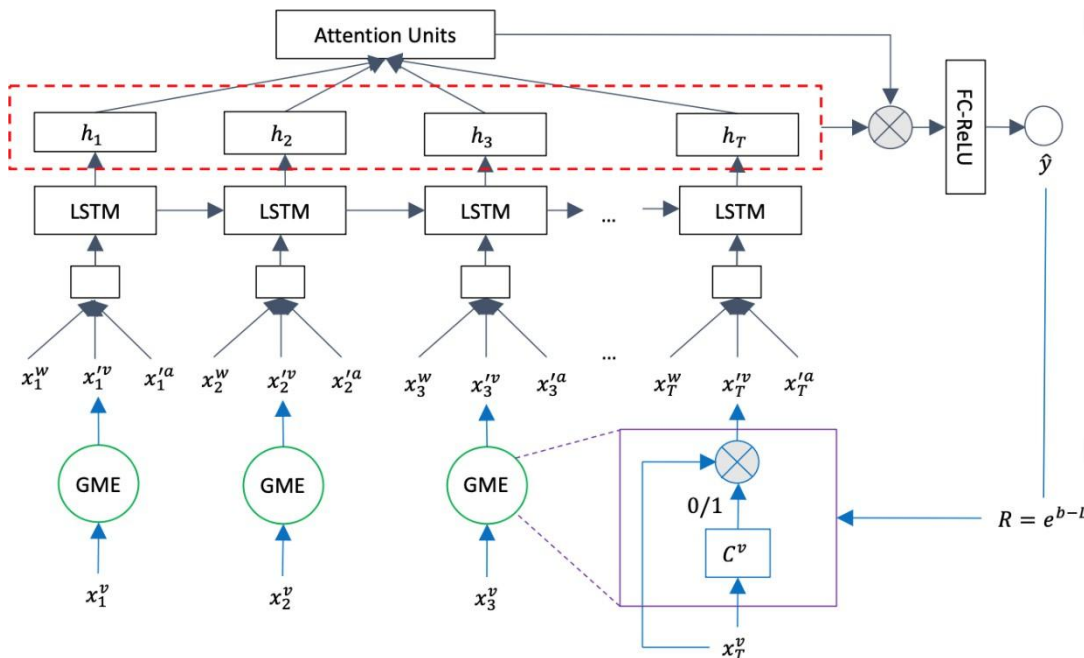
How can we build an interpretable model that estimates modality and temporal importance, and learns to attend to important information?

Minghai Chen, Sen Wang, Paul Pu Liang, Tadas Baltrušaitis, Amir Zadeh, Louis-Philippe Morency, Multimodal Sentiment Analysis with Word-Level Fusion and Reinforcement Learning, ICMI 2017, <https://arxiv.org/abs/1802.00924>

Project Example: Word-Level Gated Fusion

Solution:

- Word-level alignment
- Temporal attention over words
- Gated attention over modalities



Hypothesis: attention weights represent contribution of each modality at each time step

Modality gates that determine importance and contribution of each modality – trained with reinforcement learning

Project Example: Instruction Following

Research task: Task-Oriented Language Grounding in an Environment

Datasets: ViZDoom, based on the Doom video game

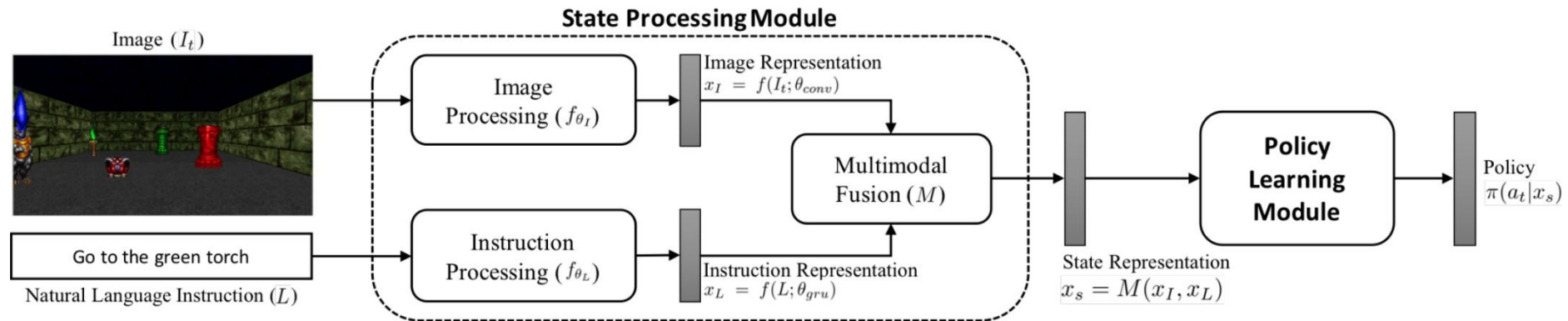
Main idea: Build a model that comprehends natural language instructions, grounds the entities and relations to the environment, and execute the instruction.



Devendra Singh Chaplot, Kanthashree Mysore Sathyendra, Rama Kumar Pasumarthi, Dheeraj Rajagopal, Ruslan Salakhutdinov, Gated-Attention Architectures for Task-Oriented Language Grounding. AAAI 2018 <https://arxiv.org/abs/1706.07230>

Project Example: Instruction Following

Solution: Gated attention architecture to attend to instruction and states



Hypothesis: Gated attention learns to ground and compose attributes in natural language with the image features. e.g. learning grounded representations for 'green' and 'torch'.

Project Example: Adversarial Attacks on VQA models

Research task: Adversarial Attacks on VQA models

Datasets: VQA

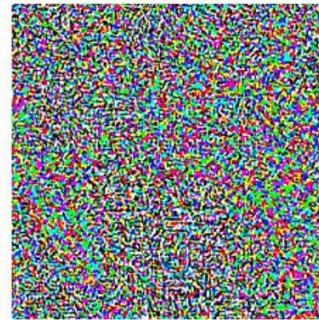
Main idea: Test the robustness of VQA models to adversarial attacks on the image.



“panda”

57.7% confidence

+ .007 ×



noise

=



“gibbon”

99.3% confidence

Project Example: Adversarial Attacks on VQA models

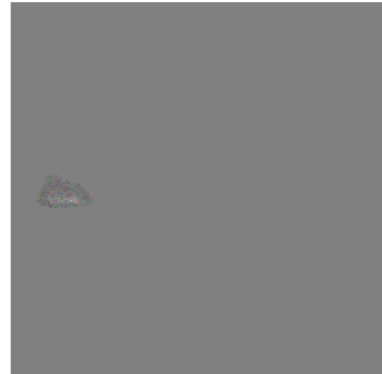
Research task: Adversarial Attacks on VQA models

Datasets: VQA

Main idea: Test the robustness of VQA models to adversarial attacks on the image.



+



Q: what kind of flowers are in the vase?



VQA model



A: **Roses** to **Sunflower**

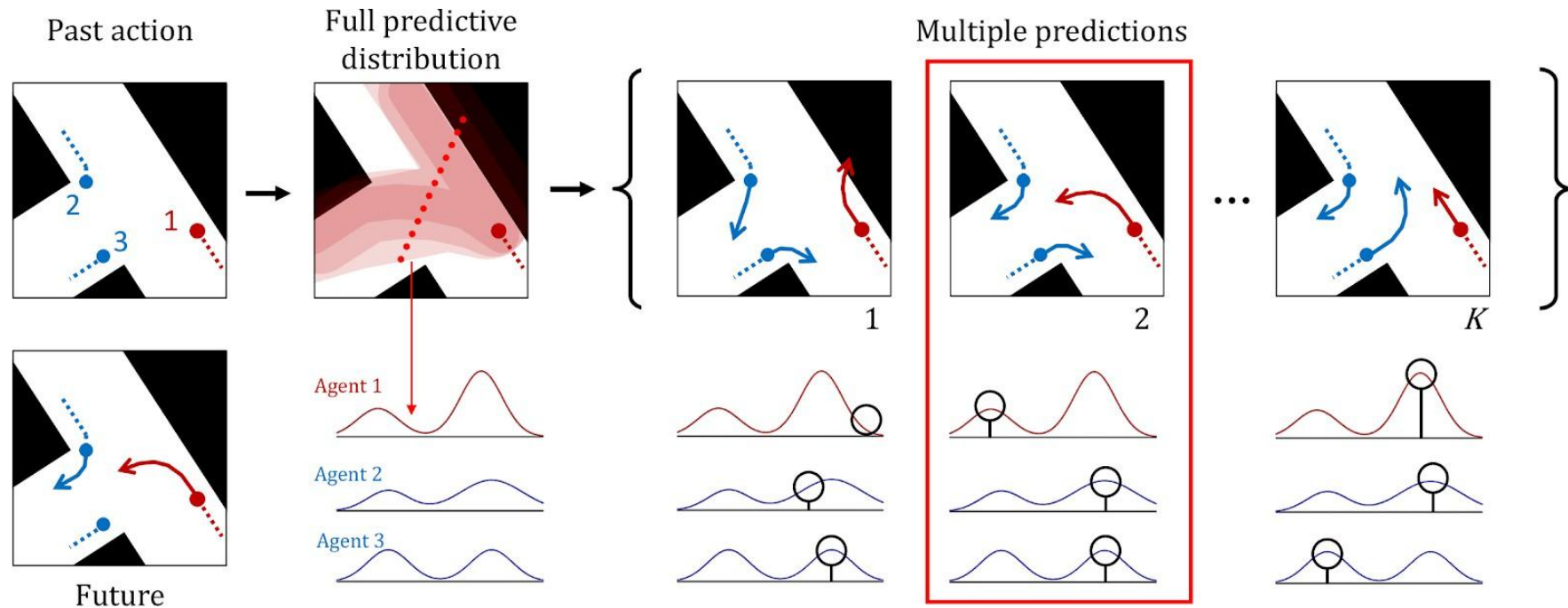
How can we design a targeted attack on images in VQA models, which will help in assessing robustness of existing models?

Project Example: Multiagent Trajectory Forecasting

Research task: Multiagent trajectory forecasting for autonomous driving

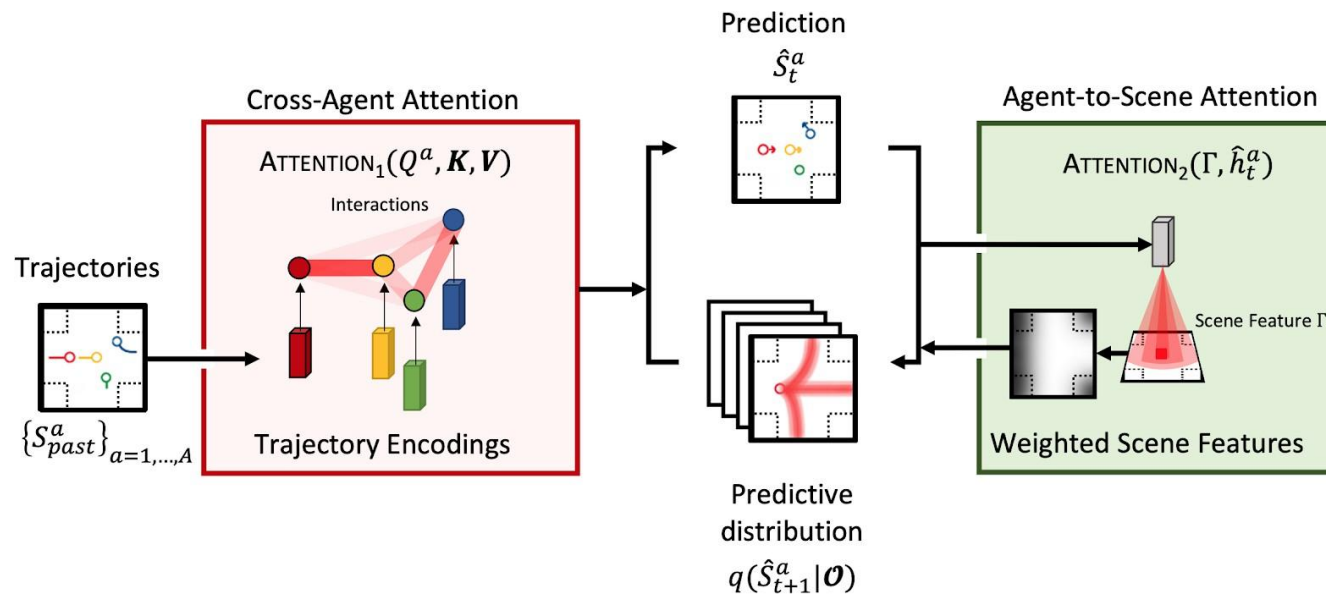
Datasets: Argoverse and Nuscenes autonomous driving datasets

Main idea: Build a model that understands the environment and multiagent trajectories and predicts a set of multimodal future trajectories for each agent.



Project Example: Multiagent Trajectory Forecasting

Solution: Modeling the environment and multiple agents to learn a distribution of future trajectories for each agent.



Hypothesis: both agent-agent interactions and agent-scene interactions are important!

Seong Hyeon Park, Gyubok Lee, Manoj Bhat, Jimin Seo, Minseok Kang, Jonathan Francis, Ashwin R. Jadhav, Paul Pu Liang, Louis-Philippe Morency, Diverse and Admissible Trajectory Forecasting through Multimodal Context Understanding. ECCV 2020
<https://arxiv.org/abs/1706.07230>

More Project Examples

See the Fall 2020 course website:

<https://cmu-multicomp-lab.github.io/mmml-course/fall2020/projects/>

